

Tutorial – 3

1. Draw a real picture for class and object. Differentiate class and object in terms of diagram only.

Perform following tasks.

Task 1: Create a class

Task 2: Add few data members as private

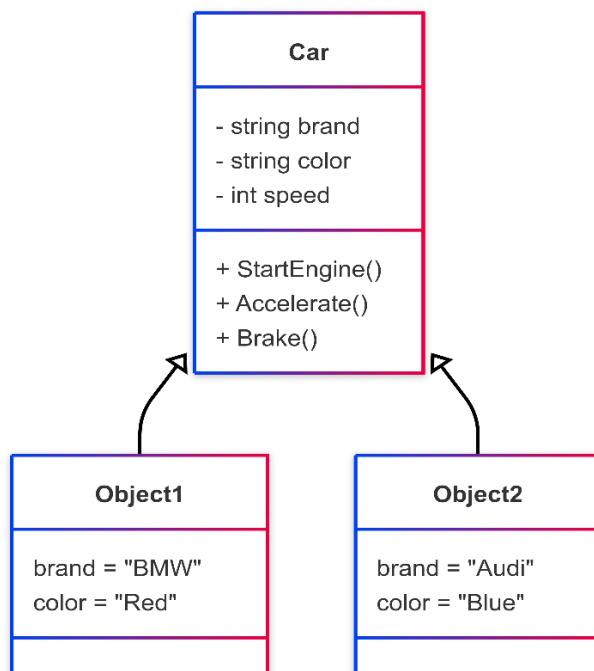
Task 3: Add few methods as public to work on defined data members

Task 4: Create a Demo class with main method.

Task 5: Create at least two objects of a class defined in Task 1 into main method and call all methods using that object.

Task 6: Write comment for each important portion of code like data members' declaration, methods, some important logic etc.

Task 7: Summarize above solution in your own few words to visualize the solution to the end user.



```
using System;

namespace Tutorial
{
    class Car           // Task 1: Create a class
    {
        // Task 2: Private Data Members

        private string brand;

        private string color;

        private int speed;

        public Car(string b, string c)      // Constructor
        {
            brand = b;

            color = c;

            speed = 0;
        }

        // Task 3: Public Methods

        public void StartEngine()           // Start Engine
        {
            Console.WriteLine(color + " " + brand + " Car is starting.");
        }
    }
}
```

```
public void Accelerate(int increment)    // Accelerate
{
    speed = speed + increment;

    Console.WriteLine(brand + " is running at " + speed + " km/h.");
}

public void Brake(int decrement)        // Brake
{
    speed = speed - decrement;

    if (speed < 0)
    {
        speed = 0;
    }

    Console.WriteLine(brand + " slowed down to " + speed + " km/h.");
}

}

// Task 4: class with Main method

class Q1
{
    static void Main(string[] args)
```

```
{  
  
    // Task 5: Create at least two objects  
  
    Car object1 = new Car("BMW", "Red");  
  
    Car object2 = new Car("Audi", "Blue");  
  
  
    object1.StartEngine();  
  
    object1.Accelerate(50);  
  
    object1.Brake(20);  
  
  
    Console.WriteLine("-----");  
  
    object2.StartEngine();  
  
    object2.Accelerate(70);  
  
    object2.Brake(40);  
  
}  
  
}  
  
}
```

Output:

```
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Red BMW Car is starting.
BMW is running at 50 km/h.
BMW slowed down to 30 km/h.
-----
Blue Audi Car is starting.
Audi is running at 70 km/h.
Audi slowed down to 30 km/h.
```

2. Define a class **Clock** with three *private* integer data members **hour**, **min** and **sec**. Define a no argument constructor to initialize time value to 12:00:00. Define a three argument constructor to initialize the time.

Define a methods to

- Increment time to next second.
- Display the time value.
- Return the hour (*int getHour()*)
- Return the minute (*int getMinute()*)
- Return the seconds (*int getSeconds()*)

using System;

namespace Tutorial

{

class Clock

{

// Private data members

private int hour;

private int min;

private int sec;

```
// No-argument constructor (default time 12:00:00)
```

```
public Clock()
```

```
{
```

```
    hour = 12;
```

```
    min = 0;
```

```
    sec = 0;
```

```
}
```

```
// Three-argument constructor
```

```
public Clock(int h, int m, int s)
```

```
{
```

```
    hour = h;
```

```
    min = m;
```

```
    sec = s;
```

```
}
```

```
// Method to increment time by one second
```

```
public void Tick()
```

```
{
```

```
    sec = sec + 1; // Increase seconds by 1
```

```
    if (sec == 60)
```

```
{
```

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```
        sec = 0;

        min = min + 1;

    }

    if (min == 60)

    {

        min = 0;

        hour = hour + 1;

    }

    if (hour == 24)

    {

        hour = 0;

    }

}

// Method to display current time (simple version)

public void DisplayTime()

{

    Console.WriteLine("Time = " + hour + ":" + min + ":" + sec);

}

// Getter methods

public int GetHour()

{
```

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```
        return hour;
    }

    public int GetMinute()
    {
        return min;
    }

    public int GetSeconds()
    {
        return sec;
    }
}

class Demo
{
    static void Main(string[] args)
    {
        // Using no-argument constructor
        Clock clock1 = new Clock();

        Console.WriteLine("Clock1 initial time:");

        clock1.DisplayTime();
    }
}
```

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```
clock1.Tick(); // increment by 1 second

Console.WriteLine("Clock1 after 1 tick:");

clock1.DisplayTime();

Console.WriteLine("-----");


// Using three-argument constructor

Clock clock2 = new Clock(23, 59, 58);

Console.WriteLine("Clock2 initial time:");

clock2.DisplayTime();


clock2.Tick(); // 23:59:58 → 23:59:59

clock2.DisplayTime();


clock2.Tick(); // 23:59:59 → 00:00:00

clock2.DisplayTime();


// Using getter methods

Console.WriteLine("Hour: " + clock2.GetHour());

Console.WriteLine("Minute: " + clock2.GetMinute());

Console.WriteLine("Second: " + clock2.GetSeconds());
```

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```
Console.ReadLine();  
  
    }  
  
    }  
  
}
```

Output :

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_3>Q2  
Clock1 initial time:  
Time = 12:0:0  
Clock1 after 1 tick:  
Time = 12:0:1  
-----  
Clock2 initial time:  
Time = 23:59:58  
Time = 23:59:59  
Time = 0:0:0  
Hour: 0  
Minute: 0  
Second: 0
```

3. Define a Student class with appropriate data members, property, constructors, methods etc. Define another class called TestStudent within the same .cs file. Also create an object of student class and demonstrate the use of student class.

```
using System;  
  
using System.Collections.Generic;  
  
using System.Linq;  
  
using System.Text;  
  
using System.Threading.Tasks;  
  
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_3  
{
```

```
// Student class

class Student
{
    // Data members (private fields)

    private int rollNo;

    private string name;

    private double marks;

    // Properties (to access private fields safely)

    public int RollNo
    {
        get { return rollNo; }

        set { rollNo = value; }
    }

    public string Name
    {
        get { return name; }

        set { name = value; }
    }

    public double Marks
    {
        get { return marks; }
```

```
        set { marks = value; }

    }

    // Default constructor

    public Student()

    {

        rollNo = 0;

        name = "Unknown";

        marks = 0.0;

    }

    // Parameterized constructor

    public Student(int rollNo, string name, double marks)

    {

        this.rollNo = rollNo;

        this.name = name;

        this.marks = marks;

    }

    // Method to display student details

    public void DisplayStudent()

    {

        Console.WriteLine("Roll No: " + rollNo);

        Console.WriteLine("Name : " + name);

    }

}
```

```
        Console.WriteLine("Marks : " + marks);  
  
    }  
  
    // Method to check result  
  
    public void CheckResult()  
  
    {  
  
        if (marks >= 40)  
  
            Console.WriteLine(name + " has Passed.");  
  
        else  
  
            Console.WriteLine(name + " has Failed.");  
  
    }  
  
}  
  
class Q3  
  
{  
  
    static void Main(string[] args)  
  
    {  
  
        // Using default constructor  
  
        Student s1 = new Student();  
  
        Console.WriteLine("Student 1 (Default:");  
  
        s1.DisplayStudent();  
  
        s1.CheckResult();  
  
    }  
  
}
```

```
Console.WriteLine("-----");  
  
// Using parameterized constructor  
  
Student s2 = new Student(101, "Saurav", 85.5);  
  
Console.WriteLine("Student 2 (Parameterized):");  
  
s2.DisplayStudent();  
  
s2.CheckResult();
```

```
Console.WriteLine("-----");
```

```
// Using properties to set values
```

```
Student s3 = new Student();  
  
s3.RollNo = 102;  
  
s3.Name = "Rahul";  
  
s3.Marks = 32.0;
```

```
Console.WriteLine("Student 3 (Using Properties):");
```

```
s3.DisplayStudent();  
  
s3.CheckResult();
```

```
}
```

```
}
```

```
}
```

Output :

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_3>Q3
Student 1 (Default):
Roll No: 0
Name   : Unknown
Marks  : 0
Unknown has Failed.
-----
Student 2 (Parameterized):
Roll No: 101
Name   : Saurav
Marks  : 85.5
Saurav has Passed.
-----
Student 3 (Using Properties):
Roll No: 102
Name   : Rahul
Marks  : 32
Rahul has Failed.
```

4. Use above program classes and create objects for 5 students and demonstrate the use student class.

```
using System;

using System.Collections.Generic;

using System.Linq;

using System.Text;

using System.Threading.Tasks;

namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_3

{

    // Student class
```

```
class Student
{
    // Data members (private fields)

    private int rollNo;

    private string name;

    private double marks;

    // Properties (to access private fields safely)

    public int RollNo
    {
        get { return rollNo; }

        set { rollNo = value; }
    }

    public string Name
    {
        get { return name; }

        set { name = value; }
    }

    public double Marks
    {
        get { return marks; }

        set { marks = value; }
```

```
}

// Default constructor

public Student()

{

    rollNo = 0;

    name = "Unknown";

    marks = 0.0;

}

// Parameterized constructor

public Student(int rollNo, string name, double marks)

{

    this.rollNo = rollNo;

    this.name = name;

    this.marks = marks;

}

// Method to display student details

public void DisplayStudent()

{

    Console.WriteLine("Roll No: " + rollNo);

    Console.WriteLine("Name : " + name);

    Console.WriteLine("Marks : " + marks);
```

```
}  
  
// Method to check result  
  
public void CheckResult()  
{  
  
    if (marks >= 40)  
  
        Console.WriteLine(name + " has Passed.");  
  
    else  
  
        Console.WriteLine(name + " has Failed.");  
  
}  
}
```

```
class Q3  
{  
  
    static void Main(string[] args)  
  
    {  
  
        // Using default constructor  
  
        Student s1 = new Student();  
  
        Console.WriteLine("Student 1 (Default:");  
  
        s1.DisplayStudent();  
  
        s1.CheckResult();  
  
    }  
}
```

```
Console.WriteLine("-----");

// Using parameterized constructor

Student s2 = new Student(101, "Saurav", 85.5);

Console.WriteLine("Student 2 (Parameterized):");

s2.DisplayStudent();

s2.CheckResult();


Console.WriteLine("-----");


// Using properties to set values

Student s3 = new Student();

s3.RollNo = 102;

s3.Name = "Rahul";

s3.Marks = 32.0;


Console.WriteLine("Student 3 (Using Properties):");

s3.DisplayStudent();

s3.CheckResult();


Console.WriteLine("-----");
```

```
// Student 4 (Parameterized constructor)
```

```
Student s4 = new Student(103, "Anita", 67.5);
```

```
Console.WriteLine("Student 4:");
```

```
s4.DisplayStudent();
```

```
s4.CheckResult();
```

```
Console.WriteLine("-----");
```

```
// Student 5 (Using properties)
```

```
Student s5 = new Student();
```

```
s5.RollNo = 104;
```

```
s5.Name = "Vikas";
```

```
s5.Marks = 45.0;
```

```
Console.WriteLine("Student 5:");
```

```
s5.DisplayStudent();
```

```
s5.CheckResult();
```

```
}
```

```
}
```

```
}
```

Output :

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_3>Q3
Student 1 (Default):
Roll No: 0
Name   : Unknown
Marks  : 0
Unknown has Failed.
-----
Student 2 (Parameterized):
Roll No: 101
Name   : Saurav
Marks  : 85.5
Saurav has Passed.
-----
Student 3 (Using Properties):
Roll No: 102
Name   : Rahul
Marks  : 32
Rahul has Failed.
-----
Student 4:
Roll No: 103
Name   : Anita
Marks  : 67.5
Anita has Passed.
-----
Student 5:
Roll No: 104
Name   : Vikas
Marks  : 45
Vikas has Passed.
```

5. Rearrange the given code to get the desired output.

```
using System;
```

```
using System.Collections.Generic;
```

```
using System.Linq;
```

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```
using System.Text;

using System.Threading.Tasks;

namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_3
{
    class Product
    {
        // Data members

        private string pcode;

        private string pname;

        private string mname;

        // Constructor

        public Product(string pcd, string pnm, string mnm)
        {
            pcode = pcd;

            pname = pnm;

            mname = mnm;
        }

        // Display method

        public void Display()
        {
            Console.WriteLine("\nProduct Code:= " + pcode);
```

```
        Console.WriteLine("Product Name:= " + pname);

        Console.WriteLine("Manufacturer Name:= " + mname);
    }
}

internal class Q5
{
    public static void Main(string[] args)
    {
        int n = args.Length;

        if (n < 3)
        {
            Console.WriteLine("Syntax Error\n");

            Console.WriteLine("Must Have THREE Arguments\n");

            Console.WriteLine("Please, Write as [csc TestProduct ProductCode ProductName\nManufacturer] \n");
        }

        else
        {
            // Take input from command-line arguments

            string pcd = args[0];
```

```
string pnm = args[1];

string mnm = args[2];

// Create Product object

Product p = new Product(pcd, pnm, mnm);

// Display product details

p.Display();

Console.ReadLine();

}

}

}

}
```

Output :

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_3>Q5 P001 Mouse Logitech

Product Code:= P001
Product Name:= Mouse
Manufacturer Name:= Logitech
```

6. Complete the following code that will generate the given output:

```
using System;

using System.Collections.Generic;

using System.Linq;

using System.Text;

using System.Threading.Tasks;
```

```
namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_3
{
    class Line
    {
        private double length; // Length of a line

        // Constructor
        public Line()
        {
            length = 10; // Missing statement-1: initialize length

            Console.WriteLine("Object is being created, length = " + length); // Missing statement-2
        }

        // Set length
        public void setLength(double len)
        {
            length = len; // Missing statement-3
        }

        // Get length
        public double getLength()
        {
            return length; // Missing statement-4
        }
    }
}
```

```
}  
  
internal class Q6  
{  
    static void Main(string[] args)  
    {  
        Line line = new Line(); // Object creation  
  
        // First display (default length 10)  
  
        Console.WriteLine("Length of line : {0}", line.getLength());  
  
        // Set new length  
  
        line.setLength(6);  
  
        // Second display (updated length 6)  
  
        Console.WriteLine("Length of line : {0}", line.getLength());  
  
        Console.ReadKey();  
    }  
}  
}
```

Output :

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_3>Q6  
Object is being created, length = 10  
Length of line : 10  
Length of line : 6
```

7. Define EnrolmentNo and Name properties for the Student class and demonstrate use of these properties along with required data members, methods and constructors.

```
using System;

using System.Collections.Generic;

using System.Linq;

using System.Text;

using System.Threading.Tasks;

namespace _24SOECE13901_SAURAV_PANDIT_Tutorial_3

{

    class student

    {

        public int EnrolmentNo;

        public string Name;

        // Parameterized constructor

        public student(int enrolmentNo, string name)

        {

            EnrolmentNo = enrolmentNo;

            Name = name;

        }

        public void Display()

        {
```

```
        Console.WriteLine("Enrolment No: " + EnrolmentNo);

        Console.WriteLine("Name      : " + Name);
    }
}

internal class Q7
{
    static void Main(string[] args)
    {
        student s1 = new student(101, "Saurav");

        student s2 = new student(102, "Rahul");

        s1.Display();

        Console.WriteLine("-----");

        s2.Display();

    }
}
```

Output :

```
D:\RKU\5th SEM\ .NET and C#\24SOECE13901_SAURAV_PANDIT_Tutorial_3>Q7
Enrolment No: 101
Name      : Saurav
-----
Enrolment No: 102
Name      : Rahul
```